

Statistics for Data Science Day 5: Relative Measures of Dispersion

Context is Everything: Comparing Data Variability with CV and CQD



DATA SCIENCE WITH SHIKHAR

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In Day 4, we covered absolute measures of spread such as range, variance, standard deviation, MAD, and IQR. These are powerful tools, but they all share a common blind spot: they give you a number with no context. Today we fix that.

The Problem with Absolute Measures

Absolute measures of dispersion give you a single value to quantify spread. A larger value means more spread, a smaller value means the data is tightly clustered. Simple enough until you try to compare two very different datasets.

Here's the problem: **is a standard deviation of 200 large or small?**

The honest answer is: *it depends entirely on the mean.*

- If the mean is **150**, a SD of 200 means the data is spread wider than its own centre indicating enormous variability.
- If the mean is **2,000,000**, a SD of 200 is practically negligible and the data is extremely tightly packed.

The absolute number 200 means something completely different in each context. This is the core limitation of absolute measures, they lack

a reference point for comparison.

Relative measures of dispersion solve this by expressing spread as a proportion of a central value, producing a unitless ratio that is meaningful regardless of the scale or units of the original data. This makes it possible to fairly compare the variability of completely different datasets such as comparing salary consistency across companies, or satisfaction score stability across products.

In this article, we cover two key relative measures:

- **Coefficient of Variation (CV)**
- **Coefficient of Quartile Deviation (CQD)**

Absolute vs. Relative Measures at a Glance

	Absolute Measures	Relative Measures
Output	Value in original units (e.g., \$, cm, points)	Unitless ratio or percentage
Useful for	Describing spread within a single dataset	Comparing spread across different datasets
Examples	Range, Variance, SD, MAD, IQR	CV, Coefficient of QD
Limitation	Cannot be fairly compared across different scales	Requires a meaningful central value to divide by

Coefficient of Variation (CV)

The Coefficient of Variation expresses the standard deviation as a **percentage of the mean**. It answers the question: *“How large is the spread relative to the average?”*

$$CV = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100\%$$

A higher CV means more variability relative to the mean. A lower CV means the data is more consistent relative to its average.

Key insight: CV is unitless. Whether you're comparing salaries in dollars, heights in centimetres, or temperatures in Celsius, the CV gives you a fair, apples-to-apples comparison.

Worked Example: Salary Consistency Across Companies

A recruiter wants to understand which company offers more consistent salaries:

	Company A	Company B
Mean Salary	\$50,000.00	\$1,60,000.00
Standard Deviation	\$5,000.00	\$10,000.00
CV	$5,000/50,000 = 10\%$	$10,000/160,000 = 6.25\%$

If you only looked at the standard deviation, you'd conclude Company B's salaries are more variable as its SD (\$10,000) is double Company A's (\$5,000). But that conclusion is **misleading**.

The CV reveals the truth: Company B's salaries are actually **more consistent** relative to their scale. A \$10,000 spread around a \$160,000 mean is far less significant than a \$5,000 spread around a \$50,000 mean.

```
def coefficient_of_variation(data):  
    import numpy as np  
    mean = np.mean(data)  
    std = np.std(data, ddof=1)  
    return (std / mean) * 100  
  
company_a = [45000, 48000, 50000, 52000, 55000]
```

```
company_b = [150000, 155000, 160000, 165000, 170000]
```

```
print(f"Company A CV: {coefficient_of_variation(company_a):.2f}%")
```

```
print(f"Company B CV: {coefficient_of_variation(company_b):.2f}%")
```

More Real-World Uses of CV

Domain	How CV is Used
Finance	Comparing risk-to-return ratio across investments with different average returns
Manufacturing	Monitoring production consistency where low CV means high quality control
Medicine	Comparing variability of lab measurements across different tests with different scales
Machine Learning	Assessing stability of model performance metrics across cross-validation folds

When NOT to Use CV

CV has two important limitations:

1. **Never use CV when the mean is zero or near zero** as dividing by zero (or near-zero) produces an undefined or meaninglessly large result.
2. **Avoid CV with data that can be negative** (e.g., profit/loss figures, temperature in Celsius) as a negative mean makes the CV sign ambiguous and hard to interpret. In such cases, use absolute measures instead.

Note : “In finance, a closely related concept is the ***Sharpe Ratio***, which measures return relative to risk which is essentially a variant of the CV applied to investment performance.”

Coefficient of Quartile Deviation (CQD)

The Coefficient of Quartile Deviation is the **relative version of the IQR**. Just as CV expresses SD relative to the mean, CQD expresses the IQR

relative to the sum of Q1 and Q3.

$$\text{CQD} = \frac{Q3 - Q1}{Q3 + Q1}$$

The result is a value between 0 and 1 (or 0% to 100% if multiplied by 100). A value closer to 0 means the middle 50% of the data is tightly clustered; a value closer to 1 means it is widely spread.

Why CQD Over CV?

Because CQD is based on quartiles rather than the mean and standard deviation, it is:

- **Robust to outliers** : extreme values don't influence Q1 or Q3
- **Valid for skewed data** : doesn't assume symmetry around the mean
- **Useful for ordinal data** : can be applied where mean-based measures aren't appropriate

This makes CQD the right tool whenever your data has outliers, is skewed, or when you specifically want to assess consistency of the core user/customer base hence ignoring extreme cases.

Worked Example: Product Satisfaction Comparison

A market research analyst wants to compare satisfaction consistency across two software products. Product A has a few extreme low scores from users who hit a rare bug and the analyst wants to understand the *core* user experience without those outliers skewing the result.

Customer satisfaction scores (scale 1–100) from 11 users each:

Product A (sorted): 25, 78, 80, 81, 84, 85, 88, 90, 91, 95, 98

Product B (sorted): 65, 68, 70, 72, 75, 79, 85, 88, 92, 96, 99

Finding Quartiles for 11 Values

For a sorted dataset of 11 values:

- **Q1** = 3rd value (the median of the lower 5 values)
- **Q3** = 9th value (the median of the upper 5 values)

	Product A	Product B
Q1 (3rd value)	80	70
Q3 (9th value)	91	92
IQR (Q3 – Q1)	11	22
Q3 + Q1	171	162
CQD	$11/171 \approx 0.064$	$22/162 \approx 0.136$

Interpretation:

Product A's CQD (0.064) is less than half of **Product B's** (0.136). This means the middle 50% of Product A's users have far more consistent satisfaction scores. Crucially, the outlier score of 25 for Product A had **zero influence** on this result as it fell outside the Q1–Q3 range entirely.

If the analyst had used standard deviation instead, Product A's outlier would have dragged its SD up significantly, potentially making Product A appear *less* consistent than Product B which would be a misleading conclusion.

```
import numpy as np
```

```
product_a = [25, 78, 80, 81, 84, 85, 88, 90, 91, 95, 98]
```

```
product_b = [65, 68, 70, 72, 75, 79, 85, 88, 92, 96, 99]
```

```
def coeff_quartile_deviation(data):
    q1 = np.percentile(data, 25)
    q3 = np.percentile(data, 75)
    return (q3 - q1) / (q3 + q1)

print(f"Product A CQD: {coeff_quartile_deviation(product_a):.3f}")
print(f"Product B CQD: {coeff_quartile_deviation(product_b):.3f}")
```

Note: As with IQR in Day 4, NumPy's percentile-based Q1/Q3 may differ slightly from the manual textbook method due to interpolation.

Different textbooks may calculate Q1 and Q3 slightly differently for small datasets. The method used here which is taking the 3rd and 9th values for $n=11$, follows the standard textbook convention of using the median of each half, excluding the overall median."

CV vs. CQD: Which One to Use?

Situation	Use
Data is clean, symmetric, no major outliers	CV
Data has outliers or is skewed	CQD
Comparing investment risk vs. return	CV
Comparing consistency of customer satisfaction scores	CQD
Mean is zero or negative	Neither CV nor standard deviation instead use IQR or MAD
You need a single consistency benchmark across many different datasets	CV (most widely used in practice)

Putting It All Together: The Full Dispersion Toolkit

Now that we've covered both absolute and relative measures, here's the complete picture of your dispersion toolkit:

Measure	Type	Outlier Sensitive?	Best For
Range	Absolute	Very	Quick first look
Variance	Absolute	Yes	Mathematical calculations
Standard Deviation	Absolute	Yes	Clean, symmetric data
MAD	Absolute	Robust	Skewed data, anomaly detection
IQR	Absolute	Robust	Skewed data, box plots
CV	Relative	Yes	Comparing datasets with different means
CQD	Relative	Robust	Comparing datasets with outliers

Key Takeaway & Practical Tip

Whenever you want to compare the spread of two or more datasets that have different means or different units, **never compare absolute measures directly**. Always switch to a relative measure.

A quick decision rule:

- Data is clean and symmetric → compare using **CV**
- Data has outliers or skew → compare using **CQD**
- Within a single dataset (no comparison needed) → absolute measures (SD, IQR) are perfectly fine.

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